# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The angle subtended by a diameter at any point on the circle is: 1

**(a)** $90^\circ$ **(b)** $120^\circ$ **(c)** $60^\circ$ **(d)** $135^\circ$

**2.** The radius of a circle of diameter $14$ cm is (in cm): 1

**(a)** 7 **(b)** 6 **(c)** 9 **(d)** 8

**3.** The perpendicular from the centre of a circle to a chord of length $10$ cm bisects it. Each half is (in cm): 1

**(a)** 7 **(b)** 5 **(c)** 4 **(d)** 6

**4.** The angle subtended by an arc at the centre is $50^\circ$. The angle it subtends at a point on the remaining part of the circle is: 1

**(a)** $35^\circ$ **(b)** $15^\circ$ **(c)** $25^\circ$ **(d)** $45^\circ$

**5.** An arc subtends $20^\circ$ at a point on the circle. At the centre it subtends: 1

**(a)** $60^\circ$ **(b)** $30^\circ$ **(c)** $40^\circ$ **(d)** $50^\circ$

**6.** $ABCD$ is a cyclic quadrilateral with $\angle A = 65^\circ$. Then $\angle C =$ 1

**(a)** $135^\circ$ **(b)** $105^\circ$ **(c)** $125^\circ$ **(d)** $115^\circ$

**7.** The longest chord of a circle is its: 1

**(a)** secant **(b)** diameter **(c)** tangent **(d)** radius

**8.** $AB$ is a diameter and $C$ is a point on the circle. Then $\angle ACB =$ 1

**(a)** $90^\circ$ **(b)** $60^\circ$ **(c)** $120^\circ$ **(d)** $135^\circ$

**9.** The radius of a circle of diameter $18$ cm is (in cm): 1

**(a)** 11 **(b)** 9 **(c)** 8 **(d)** 10

**10.** The perpendicular from the centre of a circle to a chord of length $16$ cm bisects it. Each half is (in cm): 1

**(a)** 9 **(b)** 7 **(c)** 10 **(d)** 8

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** An arc of a circle subtends $36^\circ$ at the centre. Find the angle it subtends at a point on the remaining part of the circle. 2

**12.** $ABCD$ is a cyclic quadrilateral with $\angle B = 72^\circ$. Find $\angle D$. 2

**13.** $AB$ is a diameter of a circle and $C$ is a point on it with $\angle CAB = 28°$. Find $\angle ABC$. 2

**14.** An arc of a circle subtends $84°$ at the centre. Find the angle it subtends at a point on the remaining part of the circle. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A chord of a circle of radius $5$ cm is at a distance of $3$ cm from the centre. Find the length of the chord. 3

**16.** Points $A$ and $B$ lie on a circle with centre $O$ and $\angle AOB = 48°$. A point $C$ lies on the major arc. Find $\angle ACB$ and say why $\angle ADB = \angle ACB$ for another point $D$ on the major arc. 3

**17.** $ABCD$ is a cyclic quadrilateral with $\angle A = 70°$ and $\angle B = 85°$. Find $\angle C$ and $\angle D$. 3

OR

A chord of a circle of radius $5$ cm is at a distance of $4$ cm from the centre. Find the length of the chord. 3

**18.** Points $A$ and $B$ lie on a circle with centre $O$ and $\angle AOB = 66°$. A point $C$ lies on the major arc. Find $\angle ACB$ and say why $\angle ADB = \angle ACB$ for another point $D$ on the major arc. 3

OR

$ABCD$ is a cyclic quadrilateral with $\angle A = 100°$ and $\angle B = 75°$. Find $\angle C$ and $\angle D$. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A chord of length $24$ cm is at a distance of $9$ cm from the centre of a circle. Find the radius. 5

OR

$AB$ is a diameter of a circle and $C$, $D$ lie on the circle on opposite sides of $AB$. If $\angle CAB = 25°$ and $\angle DAB = 30°$, find $\angle ABC$ and $\angle ABD$. 5

**20.** A chord of length $18$ cm is at a distance of $12$ cm from the centre of a circle. Find the radius. 5

OR

$AB$ is a diameter of a circle and $C$, $D$ lie on the circle on opposite sides of $AB$. If $\angle CAB = 35°$ and $\angle DAB = 40°$, find $\angle ABC$ and $\angle ABD$. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**

MASTERMIND